

text	text	text
Non-admitted	Admitted	Admitted but later evicted

(a) <|im_start|>user
Find the sum of all integer bases $b > 9$ for which 17_b is a divisor of 97_b .<|im_end|>
<|im_start|>assistant
<think>
This is a complex or challenging question, and it is difficult to provide a direct and correct answer. I need to think about it.

[186 tokens skipped...]

(b) So the problem states that 17_b divides 97_b , which translates to $(9b + 7)$ being divisible by $(b + 7)$ in integers. Wait, hold on, is that right? Wait, 17_b is $1*b + 7$, so yes, that's $b + 7$, and 97_b is $9*b + 7$, so $9b + 7$. So we need $(9b + 7) \bmod (b + 7) = 0$. Maybe I can use polynomial division or the remainder theorem here to simplify this divisibility condition.

[4624 tokens skipped...]

(c) Wait, the problem says "the sum of all integer bases $b > 9$ ", so sum is $21 + 49 = 70$. Hmm, but wait, hold on a second! Wait, in some contexts, when dealing with bases, do we have to consider that the base has to be greater than all digits, but the problem already states $b > 9$, and in 97_b , the largest digit is 9, so $b > 9$ ensures all digits (1, 7, 9) are less than b , so that's satisfied. So we don't have to worry about invalid digits here because the problem restricts $b > 9$, so we're good.

[60 tokens skipped...]

(d) Wait, another check: let's suppose someone thought 17_b is $1*7 + b$ or something, but no, in base b , the rightmost digit is units place (b^0), next is b^1 , so left digit is higher place. So for example, in base 10, 17 is $1*10 + 7$, so yes, left digit is b^1 , right is b^0 . So our conversion was correct.

[2919 tokens skipped...]

(e) Yep, I feel confident now that the sum is 70. I think that's the answer.
</think>

[start response...]